Assignment 1 Theoretical Exercises

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Exercise 1

Given the homogeneous points

$$\mathbf{x}_1 = \begin{pmatrix} -4 \\ 3 \\ 1 \end{pmatrix}, \qquad \mathbf{x}_2 = \begin{pmatrix} 4 \\ -5 \\ -2 \end{pmatrix}, \qquad \mathbf{x}_3 = \begin{pmatrix} -6\lambda \\ 12\lambda \\ 3\lambda \end{pmatrix}.$$

(1) *Cartesian coordinates*

Divide each point by its last coordinate:

$$\mathbf{x}_1 \mapsto \left(\tfrac{-4}{1}, \tfrac{3}{1}\right), \qquad \mathbf{x}_2 \mapsto \left(\tfrac{4}{-2}, \tfrac{-5}{-2}\right), \qquad \mathbf{x}_3 \mapsto \left(\tfrac{-6\lambda}{3\lambda}, \tfrac{12\lambda}{3\lambda}\right).$$

Answer. $\mathbf{x}_1$ maps to $(-4, 3)$, $\mathbf{x}_2$ maps to $(-2, \frac{5}{2})$, $\mathbf{x}_3$ maps to $(-2, 4)$.

(2) *Interpretation of* $\mathbf{x}_4$

$$\mathbf{x}_4 = \begin{pmatrix} 5 \\ -4 \\ 0 \end{pmatrix}.$$

Replace the last coordinate by a small number ε. The corresponding 2D point is

$$\left(\frac{5}{\varepsilon}, \frac{-4}{\varepsilon}\right).$$

As $\varepsilon \to 0$ this point goes to infinity in the direction $(5, -4)$.

Answer. $\mathbf{x}_4$ is a point at infinity (a vanishing point) in the direction $(5, -4)$.

(3) *Are* $\mathbf{x}_4$ *and* $\mathbf{x}_5$ *the same point?*

$$\mathbf{x}_5 = \begin{pmatrix} 5 \\ -3 \\ 0 \end{pmatrix},$$

By the argument in (2), $\mathbf{x}_4$ goes to infinity in the direction $(5, -4)$, while $\mathbf{x}_5$ goes to infinity in the direction $(5, -3)$. These directions are not the same.

Answer. No, $\mathbf{x}_4 \neq \mathbf{x}_5$, since they are points at infinity in different directions.

Exercise 2

(4) *Intersection of* $\mathbf{l}_1$ *and* $\mathbf{l}_2$

$$\mathbf{l}_1 = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \qquad \mathbf{l}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}.$$

A point $\mathbf{x} = (x, y, z)^T$ lies on a line $\mathbf{l}$ if $\mathbf{l}^T\mathbf{x} = 0$. The intersection must lie on both lines:

$$\begin{cases} \mathbf{l}_1^T\mathbf{x} = 0 \\ \mathbf{l}_2^T\mathbf{x} = 0 \end{cases} \implies \begin{cases} x + 3y + z = 0 \\ 2x - y + z = 0 \end{cases}$$

Set $z = 1$. Subtracting the equations gives $-x + 4y = 0$, i.e. $x = 4y$. Inserting into the second equation gives $7y = -1$, so

$$y = -\tfrac{1}{7}, \qquad x = -\tfrac{4}{7}.$$

Answer. $\mathbf{x} = \left(-\frac{4}{7}, -\frac{1}{7}, 1\right)^T$, which in $\mathbb{R}^2$ is the point $\left(-\frac{4}{7}, -\frac{1}{7}\right)$.

(5) *Intersection of* $\mathbf{l}_3$ *and* $\mathbf{l}_4$

$$\mathbf{l}_3 = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}, \qquad \mathbf{l}_4 = \begin{pmatrix} -3 \\ 0 \\ 3 \end{pmatrix}.$$

From the line equation:

$$\begin{cases} 2x + 2z = 0 \\ -3x + 3z = 0 \end{cases} \implies \begin{cases} x = -z \\ x = z \end{cases} \implies x = z = 0,$$

while y is free.

Answer. $\mathbf{x} = (0, 1, 0)^T$. This is a vanishing point: the intersection lies at infinity in the y-direction.

(6) *Line through* x_1 *and* x_2

$$x_1 = (1, 3) \Rightarrow \mathbf{x}_1 = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \qquad x_2 = (2, -1) \Rightarrow \mathbf{x}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}.$$

The line must satisfy $\mathbf{l}^T\mathbf{x}_1 = \mathbf{l}^T\mathbf{x}_2 = 0$. This is the same system as in (4), so we have already calculated it to be $\mathbf{l} = \left(-\frac{4}{7}, -\frac{1}{7}, 1\right)^T$. Check:

$$-\tfrac{4}{7} - \tfrac{3}{7} + 1 = 0, \qquad -\tfrac{8}{7} + \tfrac{1}{7} + 1 = 0.$$

This is because in $\mathbb{P}^2$ the equation is a plane through the origin that contains the points $\mathbf{x}_1$ and $\mathbf{x}_2$. When collapsing $\mathbb{P}^2$ to $\mathbb{R}^2$, the plane becomes the line between x_1 and x_2.

Answer. $\mathbf{l} = \left(-\frac{4}{7}, -\frac{1}{7}, 1\right)^T$.

Exercise 3

(7) *Why the null space gives the intersection*

$$M\mathbf{x} = [\mathbf{l}_2^T\mathbf{x}, \mathbf{l}_1^T\mathbf{x}]^T.$$

This is just the line equation for the two lines. Since we know that these must hold for the lines then the vector that gets created must be the zero vector.

Answer. A vector in the null space of M satisfies the line equation of both lines, so it is their intersection point.

(8) *Are there other solutions?*

Of course $\mathbf{x} = \mathbf{0}$ is also solution to this equation, but this does not have a representation in $\mathbb{R}^2$ and can therefore be disregarded for our case.

Answer. There are no other points in $\mathbb{P}^2$ that solve the equation, except the zero-vector.

Exercise 4

$$H = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \mathbf{x}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad \mathbf{x}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

(a) *Transformed points*

$$\mathbf{y}_1 = H\mathbf{x}_1 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \qquad \mathbf{y}_2 = H\mathbf{x}_2 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

Answer. $\mathbf{y}_1 = (2, 0, 1)^T, \quad \mathbf{y}_2 = (2, 2, 1)^T.$

(b) *Lines through the points*

Line $\mathbf{l}_1 = (a, b, c)^T$ through $\mathbf{x}_1$ and $\mathbf{x}_2$:

$$\begin{cases} \mathbf{l}_1^T\mathbf{x}_1 = 0 \\ \mathbf{l}_1^T\mathbf{x}_2 = 0 \end{cases} \implies \begin{cases} a + c = 0 \\ b + c = 0 \end{cases} \overset{c=1}{\implies} a = b = -1.$$

Line $\mathbf{l}_2 = (a, b, c)^T$ through $\mathbf{y}_1$ and $\mathbf{y}_2$:

$$\begin{cases} \mathbf{l}_2^T\mathbf{y}_1 = 0 \\ \mathbf{l}_2^T\mathbf{y}_2 = 0 \end{cases} \implies \begin{cases} 2a + c = 0 \\ 2a + 2b + c = 0 \end{cases} \overset{c=1}{\implies} a = -\tfrac{1}{2},\ b = 0.$$

Answer. $\mathbf{l}_1 = (-1, -1, 1)^T, \quad \mathbf{l}_2 = \left(-\tfrac{1}{2}, 0, 1\right)^T.$

(c) *Verify* $\mathbf{l}_2 \sim (H^{-1})^T \mathbf{l}_1$

Use $H^{-1} = \frac{1}{\det H} \operatorname{adj}(H)$ with

$$\det H = 2, \qquad \operatorname{adj}(H) = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & -2 \\ 0 & 0 & 2 \end{pmatrix} \implies H^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & -2 \\ 0 & 0 & 2 \end{pmatrix}, \quad (H^{-1})^T = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & -2 & 2 \end{pmatrix}.$$

Then

$$(H^{-1})^T \mathbf{l}_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & -2 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \sim \begin{pmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{pmatrix}.$$

Answer. $(H^{-1})^T \mathbf{l}_1 = (-1, 0, 2)^T \sim \mathbf{l}_2$, so the two lines are the same up to scale.

Exercise 5

Lines transform with $(H^{-1})^T$

Assume $\mathbf{x}$ lies on $\mathbf{l}_1$ and let $\mathbf{y} \sim H\mathbf{x}$. Then

$$\mathbf{l}_1^T \mathbf{x} = 0 \implies \mathbf{l}_1^T H^{-1} \underbrace{H\mathbf{x}}_{\mathbf{y}} = 0 \implies \mathbf{l}_1^T H^{-1} \mathbf{y} = \left((H^{-1})^T \mathbf{l}_1\right)^T \mathbf{y} = 0.$$

Therefore, if we set $\mathbf{l}_2 \sim (H^{-1})^T \mathbf{l}_1$, the transformed point $\mathbf{y} \sim H\mathbf{x}$ belongs to $\mathbf{l}_2$. We can always create the inverse of H because a projective transformation is always reversible.

Answer. $\mathbf{l}_2 \sim (H^{-1})^T \mathbf{l}_1$.

Exercise 6

$$H_1 = \begin{pmatrix} 1 & 1 & \sqrt{2} \\ -1 & 1 & 1 \\ 0 & 0 & \sqrt{2} \end{pmatrix}, H_2 = \begin{pmatrix} 2 & 2 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}, H_3 = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, H_4 = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & 1 \\ 1 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

(a) *Which are projective transformations?*

Answer. H_1, H_2 and H_4 are projective transformations.

(b) *Which are affine transformations?*

Answer. H_1 and H_4 are affine.

(c) *Which are similarity transformations?*

rotation matrix.

Answer. H_1.

(d) *Which are Euclidean transformations?*

Answer. H_1.

(e) *Which of the transformations preserve lengths between points?*

Answer. Only H_1 (Euclidean).

(f) *Which map lines to lines?*

Answer. H_1, H_2 and H_4

(g) *Which map parallel lines to parallel lines?*

Answer. H_1 and H_4.

Exercise 7

The camera equation is $\lambda \mathbf{x} = P\mathbf{X}$, with

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}, \qquad \mathbf{X}_1 = \begin{pmatrix} 1 \\ -1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} -2 \\ 2 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{X}_3 = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 1 \end{pmatrix}.$$

(a) *Projections*

$$P\mathbf{X}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \qquad P\mathbf{X}_2 = \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix}, \qquad P\mathbf{X}_3 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

The last coordinate of $P\mathbf{X}_2$ is 0, so its projection is a vanishing point.

Answer. $\mathbf{x}_1 \sim (1, -1, 1)^T$, $\quad \mathbf{x}_2 \sim (-2, 2, 0)^T$ (vanishing point), $\quad \mathbf{x}_3 \sim (1, 2, 2)^T$.

(b) *Camera center*

The camera center is the point that projects to $\mathbf{0}$:

$$\begin{pmatrix}0\\0\\0\end{pmatrix} = \begin{pmatrix}X_1\\X_2\\X_3-1\end{pmatrix} \implies X_1=0,\ X_2=0,\ X_3=1.$$

Answer. The camera center is $C = (0,\ 0,\ 1)$.

(c) *Principal axis*

The points on the principal axis project to the image center $(0,0,1)^T$:

$$\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}X_1\\X_2\\X_3-1\end{pmatrix} \implies X_1=0,\ X_2=0,\ X_3=2.$$

So the axis goes through $C = (0,0,1)$ and $(0,0,2)$.

Answer. The principal axis is the line through $(0,0,1)$ with direction $(0,\ 0,\ 1)$.

Exercise 8

Let $\mathbf{x} \in \mathbb{P}^2$ and $P_1 = [I\ \mathbf{0}]$.

(a) *Points that project to* $\mathbf{x}$

$$\mathbf{x} \sim P_1\mathbf{U} = [I\ \mathbf{0}]\,\mathbf{U}.$$

Write $\mathbf{U} = (x_1',\ x_2',\ x_3',\ s')^T$. Then

$$P_1\mathbf{U} = \begin{pmatrix}x_1'\\x_2'\\x_3'\end{pmatrix} = \begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} \implies \mathbf{U} \sim \begin{pmatrix}\mathbf{x}\\s\end{pmatrix}.$$

The $\mathbf{U}$ is a collection of points that creates a line in 3 dimensions. It's not possible to know the value of s from the information of P_1 because the part that multiplies with s is zero in this matrix.

Answer. $\mathbf{U}(s) = (\mathbf{x}^T,\ s)^T$ projects to $\mathbf{x}$ for every $s \in \mathbb{R}$, and this is a collection of points in 3 dimensions that creates a line. It's not possible to know the value of s by the matrix P_1.

(b) *The* s *that puts* $\mathbf{U}(s)$ *in the plane*

Assume $\mathbf{U}$ belongs to the plane $\Pi = (\pi^T,\ 1)^T$:

$$\Pi^T\mathbf{U}(s) = 0 \implies \pi^T\mathbf{x} + s = 0 \implies s = -\pi^T\mathbf{x}.$$

Answer. $s = -\pi^T\mathbf{x}$, i.e. $\mathbf{U}(s) = \begin{pmatrix}\mathbf{x}\\-\pi^T\mathbf{x}\end{pmatrix}$.

(c) *The homography* $H = R - \mathbf{t}\pi^T$

Given $\mathbf{x} \sim P_1\mathbf{U}$, $\mathbf{y} \sim P_2\mathbf{U}$ with $P_2 = [R\ \mathbf{t}]$, and $\Pi^T\mathbf{U} = 0$:

$$\mathbf{y} \sim P_2\mathbf{U}(s) = [R\ \mathbf{t}] \begin{pmatrix} \mathbf{x} \\ -\pi^T\mathbf{x} \end{pmatrix} = R\mathbf{x} - \mathbf{t}\pi^T\mathbf{x} = \left(R - \mathbf{t}\pi^T\right)\mathbf{x}.$$

Answer. $\mathbf{y} \sim H\mathbf{x}$ with $H = R - \mathbf{t}\pi^T$, so H maps $\mathbf{x}$ to $\mathbf{y}$. □